6G5Z0050 : Maths for Data Science

DURATION : 2 hours

Instructions to candidates:

Answer all the questions.

You must show all working to gain full marks.

Permitted materials:

Calculators are permitted. You may bring one double-sided A4 sheet of notes into the exam with you. A standard formula sheet will also be provided.

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION VENUE

1. (a) (i) Consider the vectors $\mathbf{u} = (1, 0, 2)$, $\mathbf{v} = (1, 3, 4)$ and $\mathbf{w} = (-3, 0, -2)$. Calculate the dot product $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w})$. (3 marks)

(ii) What would be true of two vectors whose dot product:

- is large and positive?
- is large and negative?
- is zero?

(3 marks)

(b) Consider the following system of linear equations:

$$\begin{aligned} x + 2y &= 1, \\ 4x + 9y + z &= -2, \\ 3x + 6y + z &= 2. \end{aligned}$$

(i) Present the system of equations in the matrix form $A\mathbf{x} = \mathbf{b}$. Clearly denote the values of A, $\mathbf{b}$ and $\mathbf{x}$. (5 marks)

(ii) Using Gauss-Jordan elimination, find the inverse of the matrix A from part (i). (10 marks)

(iii) Using this inverse matrix, and the vector $\mathbf{b}$ from part (i), find the solution to the system of equations. (5 marks)

Total marks for question 1: [26]

2. (a) A proof-reader reads through a 250-page proof of a book. The numbers of mistakes found on each page are summarized in the table.

Number of mistakes	0	1	2	3	4
Number of pages	61	109	53	23	4

(i) Determine the mean number of mistakes found per page. (4 marks)

(ii) Find the variance in the number of mistakes found per page. (5 marks)

(iii) If this book is opened at a random page, what is the probability that page will contain no mistakes? (2 marks)

(b) The publisher wants a report on how many mistakes are made in their books generally, and asks the proof-reader to look at 100 different book proofs. The proof-reader suggests it will take too long to check them all, but that they could check a sample of pages instead.

(i) What is a proportional stratified sample? (2 marks)

(ii) Describe how you might take a proportional stratified sample of the pages in the books. Include details of how you might use software to do this. (6 marks)

Total marks for question 2: [19]

3. (a) A scientist wishes to do a study comparing the strength people have in their left and right arms. The participants are recruited online, and asked to visit their local gym and complete as many bicep curls as they can using a heavy weight, first with one arm and then the other. They submit the number of curls they completed with each arm online.

   (i) Comment on two potential sources of inaccuracy in this data. (4 marks)

   A sample of the data reported by the participants is given below.

Left	14	15	26	15	20	23	21	7	11	25	7	9	25
Right	23	16	28	19	19	24	32	12	13	33	14	15	28

   (ii) Assuming the differences between the left and right arms follow a normal distribution, perform a paired t-test at the 5% significance level to compare the left and right arm data from the participants. (15 marks)

(b) The regression line for this data is given by

$$y = 5.91 + 0.913x$$

The PMCC is 0.876.

   (i) Based on the equation for the regression line and the value of the PMCC, describe the type of correlation observed. (3 marks)

   (ii) Calculate the residual for the first participant in the study. (3 marks)

   (iii) Use the regression line to predict how many reps someone could do with their right arm if they could do 18 with their left arm. (2 marks)

Total marks for question 3: [27]

4. (a) An ice cream vendor sells two flavours of ice cream, and for each flavour, the number of sales each day over the season is normally distributed. Let C represent the sales of chocolate ice cream, and V the sales of vanilla ice cream. The vendor has observed that

$$C \sim N(85, 11^2) \qquad \text{and} \qquad V \sim N(93, 7^2)$$

   (i) Write down a model for the total number of ice creams sold in a day, $C + V$, and explain why it is valid. (3 marks)

   (ii) What is probability that more than 200 ice creams are sold in total on a given day? (5 marks)

   (iii) The price of a chocolate ice cream is £2.80, and a vanilla ice cream is £1.90. Write down a model X for the total money taken in pounds, and explain why it's a valid model. Explain what the model tells you about how much money the vendor might expect to take in a day, using the standard deviation to make a prediction about how that might vary. (9 marks)

(b) The vendor is thinking about introducing a third flavour, and while they don't know what the mean sales for it will be, they have data of a sample of two weeks of daily sales from another vendor, which has a mean of 76 and an SD of 12.1. Give a 95% confidence interval for what the mean overall sales would be for this new flavour. (6 marks)

Total marks for question 4: [23]

1 Reference - Statistical Tables

1.1 Chi-Squared Test Probability Table

DF	1.00	0.99	0.98	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.03	0.01	0.01
1	0.0000	0.0002	0.0010	0.0039	0.0158	0.1015	0.4549	1.3233	2.7055	3.8415	5.0239	6.6349	7.8794
2	0.0100	0.0201	0.0506	0.1026	0.2107	0.5754	1.3863	2.7726	4.6052	5.9915	7.3778	9.2103	10.597
3	0.0717	0.1148	0.2158	0.3519	0.5844	1.2125	2.3660	4.1083	6.2514	7.8147	9.3484	11.345	12.838
4	0.2070	0.2971	0.4844	0.7107	1.0636	1.9226	3.3567	5.3853	7.7794	9.4877	11.143	13.277	14.860
5	0.4117	0.5543	0.8312	1.1455	1.6103	2.6746	4.3515	6.6257	9.2364	11.071	12.833	15.086	16.750
6	0.6757	0.8721	1.2373	1.6354	2.2041	3.4546	5.3481	7.8408	10.645	12.592	14.449	16.812	18.548
7	0.9893	1.2390	1.6899	2.1674	2.8331	4.2549	6.3458	9.0372	12.017	14.067	16.013	18.475	20.278
8	1.3444	1.6465	2.1797	2.7326	3.4895	5.0706	7.3441	10.219	13.362	15.507	17.535	20.090	21.955
9	1.7349	2.0879	2.7004	3.3251	4.1682	5.8988	8.3428	11.389	14.684	16.919	19.023	21.666	23.589
10	2.1559	2.5582	3.2470	3.9403	4.8652	6.7372	9.3418	12.549	15.987	18.307	20.483	23.209	25.188
11	2.6032	3.0535	3.8158	4.5748	5.5778	7.5841	10.341	13.701	17.275	19.675	21.920	24.725	26.757
12	3.0738	3.5706	4.4038	5.2260	6.3038	8.4384	11.340	14.845	18.549	21.026	23.337	26.217	28.300
13	3.5650	4.1069	5.0088	5.8919	7.0415	9.2991	12.340	15.984	19.812	22.362	24.736	27.688	29.819
14	4.0747	4.6604	5.6287	6.5706	7.7895	10.165	13.339	17.117	21.064	23.685	26.119	29.141	31.319
15	4.6009	5.2294	6.2621	7.2609	8.5468	11.037	14.339	18.245	22.307	24.996	27.488	30.578	32.801
16	5.1422	5.8122	6.9077	7.9617	9.3122	11.912	15.339	19.369	23.542	26.296	28.845	32.000	34.267
17	5.6972	6.4078	7.5642	8.6718	10.085	12.792	16.338	20.489	24.769	27.587	30.191	33.409	35.718
18	6.2648	7.0149	8.2308	9.3905	10.865	13.675	17.338	21.605	25.989	28.869	31.526	34.805	37.156
19	6.8440	7.6327	8.9065	10.117	11.651	14.562	18.338	22.718	27.204	30.144	32.852	36.191	38.582
20	7.4338	8.2604	9.5908	10.851	12.443	15.452	19.337	23.828	28.412	31.410	34.170	37.566	39.997
21	8.0337	8.8972	10.283	11.591	13.240	16.344	20.337	24.935	29.615	32.671	35.479	38.932	41.401
22	8.6427	9.5425	10.982	12.338	14.041	17.240	21.337	26.039	30.813	33.924	36.781	40.289	42.796
23	9.2604	10.196	11.689	13.091	14.848	18.137	22.337	27.141	32.007	35.172	38.076	41.638	44.181
24	9.8862	10.856	12.401	13.848	15.659	19.037	23.337	28.241	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	19.939	24.337	29.339	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	20.843	25.336	30.435	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	21.749	26.336	31.528	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	22.657	27.336	32.620	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	23.567	28.336	33.711	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	24.478	29.336	34.800	40.256	43.773	46.979	50.892	53.672

1.2 T-Test Probability Table

1-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.03	0.01	0.005	0.001	0.0005
2-tail	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.010**	**0.002**	**0.001**
1	0.000	1.000	1.376	1.963	3.078	6.314	12.710	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300

1.3 Normal Distribution - Probability Table

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999							